

IMO 2026 — Problem 2

Solution by Kimi K3

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Source: `results/kimi-k3/problem-02/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC , respectively. Let points K and L be chosen inside triangles BMC and BNC , respectively, such that K lies inside the angle LBA , L lies inside the angle ACK , and $\angle KBA = \angle ACL$, $\angle LBK = \angle LNC$, $\angle LCK = \angle BMK$. Let O be the circumcentre of triangle AKL . Prove that $OM = ON$.

Solution document

Status

solved

Approaches tried

- **Numerical exploration** (`code/explore.py`): confirmed $OM = ON$ across many triangles and parameter choices; confirmed the configuration exists and is (generically) unique for each φ .
- **Power-of-a-point reformulation:** $OM = ON \iff \text{pow}(M) = \text{pow}(N) \iff \text{pow}(B) - \text{pow}(C) = \frac{c^2 - b^2}{2}$ (median/Apollonius identity). This is the route that worked.
- **Angle chasing:** found $\cot \angle BAK = \cot \varphi + 2 \cot \gamma$, $\cot \angle CAL = \cot \varphi + 2 \cot \beta$; found $\angle MKC = \angle NLB = 180^\circ - A - \varphi$ (automatic, not the crux).
- **Law of sines in four triangles** (BMK , CNL , BKC , BCL) $\rightarrow$ two quadratic constraints ConsP , ConsQ on $p = BK$, $q = CL$. **This is the key structural fact.**
- **Coordinate computation of powers** with A at origin (complex notation $K = c - pe^{-i\varphi}$, $L = e^{iA}(b - qe^{i\varphi})$) $\rightarrow$ explicit formulas for $\text{pow}(B)$, $\text{pow}(C)$; discovered they simplify to closed forms using ConsP , ConsQ .
- **Symbolic verification** (`code/verify_identity.py`, `code/verify_power_formulas.py`): the two crucial polynomial identities are exactly true (difference simplifies to 0); the elimination R1 R3 ConsP verified exactly.

Current best

Complete rigorous proof. The crux: the given angle conditions force $p = BK$ and $q = CL$ to satisfy the quadratics $2\sin(A + \varphi)p^2 - (2\sin A + \sin(A + 2\varphi))cp + c^2\sin(A + \varphi) - bc\sin\varphi = 0$ (and the analogous one for q with b), which in turn force $\text{pow}(B) = \frac{c^2\sin(A + \varphi) - bc\sin\varphi}{2\sin(A + \varphi)}$ and $\text{pow}(C) = \frac{b^2\sin(A + \varphi) - bc\sin\varphi}{2\sin(A + \varphi)}$, so $\text{pow}(B) - \text{pow}(C) = \frac{c^2 - b^2}{2}$, equivalent to $OM = ON$.

Full proof**Notation**

Let A, B, C also denote the angles of $\triangle ABC$, and $a = BC$, $b = CA$, $c = AB$. Put

$$\varphi := \angle KBA = \angle ACL, \quad \beta := \angle LBK = \angle LNC, \quad \gamma := \angle LCK = \angle BMK,$$

and $p := BK$, $q := CL$. Since K lies inside $\angle LBA$, $\angle ABL = \beta + \varphi$; since L lies inside $\angle ACK$, $\angle ACK = \gamma + \varphi$. All angles appearing below are positive because $K \in \triangle BMC$ and $L \in \triangle BNC$ are interior points (so rays BK, BL lie inside $\angle ABC$ and rays CK, CL inside $\angle ACB$); in particular $0 < \varphi < B$, $\beta + \varphi < B$, $\gamma + \varphi < C$, whence $A + \varphi < A + B < \pi$ and $\sin(A + \varphi) > 0$.

Step 1: Reduction to powers of B and C

Let R be the radius of the circumcircle $\omega = (AKL)$ and write $\text{pow}(\cdot)$ for power w.r.t. ω . Since M is the midpoint of AB , Apollonius's median formula in $\triangle OAB$ gives

$$OM^2 = \frac{2OA^2 + 2OB^2 - AB^2}{4} = \frac{2R^2 + 2OB^2 - c^2}{4},$$

so $\text{pow}(M) = OM^2 - R^2 = \frac{\text{pow}(B)}{2} - \frac{c^2}{4}$. Similarly $\text{pow}(N) = \frac{\text{pow}(C)}{2} - \frac{b^2}{4}$. Therefore

$$OM = ON \iff \text{pow}(M) = \text{pow}(N) \iff \text{pow}(B) - \text{pow}(C) = \frac{c^2 - b^2}{2}. \quad (1)$$

Step 2: Four law-of-sines relations

Triangle BMK: angles φ at B (as $M \in BA$), γ at M , $\pi - \varphi - \gamma$ at K ; with $BM = c/2$:

$$p = BK = \frac{c \sin \gamma}{2 \sin(\varphi + \gamma)}. \quad (R1)$$

Triangle CNL: angles φ at C (as $N \in CA$), β at N , $\pi - \varphi - \beta$ at L ; with $CN = b/2$:

$$q = CL = \frac{b \sin \beta}{2 \sin(\varphi + \beta)}. \quad (R2)$$

Triangle BKC: $\angle KBC = B - \varphi$, $\angle KCB = C - (\gamma + \varphi)$ (using $\angle ACK = \gamma + \varphi$), $\angle BKC = A + \gamma + 2\varphi$:

$$p = BK = \frac{a \sin(C - \varphi - \gamma)}{\sin(A + \gamma + 2\varphi)}. \quad (R3)$$

Triangle BCL: $\angle LBC = B - (\beta + \varphi)$ (using $\angle ABL = \beta + \varphi$), $\angle BCL = C - \varphi$, $\angle BLC = A + \beta + 2\varphi$:

$$q = CL = \frac{a \sin(B - \varphi - \beta)}{\sin(A + \beta + 2\varphi)}. \quad (R4)$$

Step 3: The quadratic constraints (elimination of β, γ)

Auxiliary identity: for any x , $c \sin(A + x) = b \sin x + a \sin(C - x)$. Indeed with $a = 2R' \sin A$, $b = 2R' \sin B$, $c = 2R' \sin C$ this is equivalent to $\sin C \sin(A + x) = \sin B \sin x + \sin A \sin(C - x)$, and using $\sin B = \sin(A + C)$ the RHS equals $\sin C(\sin x \cos A + \sin A \cos x) = \sin C \sin(A + x)$.

Combining (R3) with the identity at $x = \gamma + \varphi$:

$$p \sin(A + \gamma + 2\varphi) = c \sin(A + \gamma + \varphi) - b \sin(\gamma + \varphi). \quad (\text{EqC})$$

Expanding (R1) and (EqC) with the addition formulas:

$$2p \sin \varphi \cos \gamma = (c - 2p \cos \varphi) \sin \gamma, \quad (i)$$

$$[c \cos(A + \varphi) - p \cos(A + 2\varphi) - b \cos \varphi] \sin \gamma = [p \sin(A + 2\varphi) - c \sin(A + \varphi) + b \sin \varphi] \cos \gamma. \quad (ii)$$

Eliminating $\sin \gamma, \cos \gamma$ from (i),(ii) (multiply (i) by the bracket in (ii)'s LHS and (ii) by $2p \sin \varphi$, subtract — legitimate since the resulting identity $(c - 2p \cos \varphi)(\dots) - (2p \sin \varphi)(\dots) = 0$ follows from (i) (ii)):

$$(c - 2p \cos \varphi)[p \sin(A + 2\varphi) - c \sin(A + \varphi) + b \sin \varphi] = 2p \sin \varphi[c \cos(A + \varphi) - p \cos(A + 2\varphi) - b \cos \varphi].$$

Expand. The terms $-2pb \sin \varphi \cos \varphi$ cancel on both sides. Using $2 \sin \varphi \cos(A + \varphi) = \sin(A + 2\varphi) - \sin A$, $2 \cos \varphi \sin(A + \varphi) = \sin(A + 2\varphi) + \sin A$, and $\cos \varphi \sin(A + 2\varphi) - \sin \varphi \cos(A + 2\varphi) = \sin(A + \varphi)$, everything simplifies to:

$$\boxed{2 \sin(A + \varphi) p^2 - (2 \sin A + \sin(A + 2\varphi))cp + c^2 \sin(A + \varphi) - bc \sin \varphi = 0} \quad (\text{ConsP})$$

The symmetric computation with (R2),(R4) (swap $b \leftrightarrow c$, $p \leftrightarrow q$, $\gamma \leftrightarrow \beta$) gives:

$$\boxed{2 \sin(A + \varphi) q^2 - (2 \sin A + \sin(A + 2\varphi))bq + b^2 \sin(A + \varphi) - bc \sin \varphi = 0} \quad (\text{ConsQ})$$

Step 4: Powers of B and C in coordinates

Place $A = (0, 0)$, $B = (c, 0)$, $C = (b \cos A, b \sin A)$. From $\angle KBA = \varphi$, $BK = p$ and from $\angle ACL = \varphi$, $CL = q$:

$$K = (c - p \cos \varphi, p \sin \varphi), \quad L = (b \cos A - q \cos(A + \varphi), b \sin A - q \sin(A + \varphi)).$$

The circle ω passes through the origin, so it has equation $x^2 + y^2 + ux + vy = 0$; the power of a point P equals $g(P) = x_P^2 + y_P^2 + ux_P + vy_P$. The conditions $g(K) = g(L) = 0$ form a 2×2 linear system in u, v with determinant

$$\Delta = K_x L_y - K_y L_x = bc \sin A - (bp + cq) \sin(A + \varphi) + pq \sin(A + 2\varphi) = 2[AKL] \neq 0$$

(A, K, L are not collinear since ω is the circumcircle of a genuine triangle). Solving:

$$u = \frac{|L|^2 K_y - |K|^2 L_y}{\Delta}, \quad v = \frac{|K|^2 L_x - |L|^2 K_x}{\Delta},$$

where $|K|^2 = c^2 - 2cp \cos \varphi + p^2$ and $|L|^2 = b^2 - 2bq \cos \varphi + q^2$. Therefore

$$\text{pow}(B) = c^2 + cu = c^2 + \frac{c[|L|^2 p \sin \varphi - |K|^2 (b \sin A - q \sin(A + \varphi))]}{\Delta}, \quad (2)$$

and, using $L_x \sin A - L_y \cos A = q \sin \varphi$ and $K_y \cos A - K_x \sin A = p \sin(A + \varphi) - c \sin A$,

$$\text{pow}(C) = b^2 + ub \cos A + vb \sin A = b^2 + \frac{b[|K|^2 q \sin \varphi + |L|^2 (p \sin(A + \varphi) - c \sin A)]}{\Delta}. \quad (3)$$

Step 5: Key algebraic identities

Write $s_1 = \sin(A + \varphi)$, $s_\varphi = \sin \varphi$, $s_A = \sin A$. The following two identities hold, where each side is a polynomial in p, q whose coefficients are trigonometric polynomials in A, φ :

$$2s_1 \Delta \left(\text{pow}(B) - \frac{c^2 s_1 - bcs_\varphi}{2s_1} \right) = c(s_1 q - bs_A) \cdot \text{ConsP} + cs_\varphi p \cdot \text{ConsQ}, \quad (4)$$

$$2s_1 \Delta \left(\text{pow}(C) - \frac{b^2 s_1 - bcs_\varphi}{2s_1} \right) = bs_\varphi q \cdot \text{ConsP} + b(s_1 p - cs_A) \cdot \text{ConsQ}, \quad (5)$$

where $\text{pow}(B), \text{pow}(C)$ are substituted from (2),(3) and $\text{ConsP}, \text{ConsQ}$ from Step 3.

Verification of (4). Multiplying (2) out, the LHS of (4) equals

$$\Delta(c^2 s_1 + bcs_\varphi) + 2s_1 c \left[|L|^2 ps_\varphi - |K|^2 (bs_A - qs_1) \right].$$

Both this and the RHS of (4) are polynomials in p, q involving only the monomials $p^2 q, pq^2, p^2, q^2, pq, p, q, 1$. Comparing coefficients monomial by monomial, the two sides agree immediately in every monomial except possibly pq ; the difference of the two pq -coefficients is $c \cdot E(A, \varphi)$ with

$$E = b[3s_A s_\varphi + 4s_\varphi^2 \cos(A + \varphi) - \cos A \cos \varphi + \cos(A + 3\varphi)] + 4c[s_A s_1 + \cos A \cos(A + \varphi) - \cos \varphi].$$

The second bracket is $\cos((A + \varphi) - A) - \cos \varphi = 0$. For the first, write $\cos(A + 3\varphi) = \cos(A + \varphi) \cos 2\varphi - \sin(A + \varphi) \sin 2\varphi$; then

$$\begin{aligned} s_\varphi^{-1} \cdot (\text{first bracket}) &= 3s_A s_\varphi + 4s_\varphi^2 \cos(A + \varphi) - \cos A \cos \varphi + \cos(A + \varphi)(1 - 2s_\varphi^2) - 2s_1 s_\varphi \cos \varphi \\ &= 3s_A s_\varphi - \cos A \cos \varphi + \cos(A + \varphi) + 2s_\varphi [s_\varphi \cos(A + \varphi) - s_1 \cos \varphi] \\ &= 3s_A s_\varphi - \cos A \cos \varphi + \cos(A + \varphi) + 2s_\varphi \sin(\varphi - (A + \varphi)) \\ &= s_A s_\varphi - \cos A \cos \varphi + \cos(A + \varphi) = -\cos(A + \varphi) + \cos(A + \varphi) = 0. \end{aligned}$$

Hence $E = 0$ and (4) holds. (All steps are direct expansions using only the addition formulas; the computation was also verified symbolically in `code/verify_power_formulas.py`.) Identity (5) is the same computation with the roles $b \leftrightarrow c$, $p \leftrightarrow q$ interchanged.

Step 6: Conclusion

Since $\text{ConsP} = \text{ConsQ} = 0$ (Step 3) and $s_1 \Delta \neq 0$, identities (4),(5) give

$$\text{pow}(B) = \frac{c^2 \sin(A + \varphi) - bc \sin \varphi}{2 \sin(A + \varphi)}, \quad \text{pow}(C) = \frac{b^2 \sin(A + \varphi) - bc \sin \varphi}{2 \sin(A + \varphi)}.$$

Subtracting:

$$\text{pow}(B) - \text{pow}(C) = \frac{(c^2 - b^2) \sin(A + \varphi)}{2 \sin(A + \varphi)} = \frac{c^2 - b^2}{2}.$$

By (1) this is equivalent to $OM = ON$. ■